Royal Devon and Exeter NHS
NHS Foundation Trust

LocSSIP Guideline for: Invasive Procedures in Critical Care

SUMMARY

This guideline outlines the process to be followed for invasive procedures in Critical Care (excluding operating theatres). It applies to all patients undergoing the following invasive procedures.

- Insertion of a Central Venous Catheter
- Insertion of a Chest Drain
- Emergency Intubation
- Percutaneous Tracheostomy

The guideline has been written as a result of National Safety Standards for Invasive Procedures (NatSSIPs), **national guidance which requires detailed management plans for all invasive procedures undertaken in an outpatient, treatment room or ward setting.**

PUBLICATION DETAILS

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1. INTRODUCTION

1.1 The Royal Devon and Exeter NHS Foundation Trust (the Trust) is committed to providing high quality patient centred care and a safe environment for all patients undergoing any invasive procedure.

2. BACKGROUND

2.1 An NHS England Patient Safety Alert (September 2015) recognises that 'surgical' Never Events are relevant to all clinical settings where invasive procedures are undertaken and requires Trusts to broaden the scope of evidence-based standards that build on the WHO Surgical Safety Checklist approach in order to address many of the underlying causes of Never Events.

2.2 This guideline is intended to build on the principles outlined in the Trust's Safe Surgery and Interventional Procedures Policy and ensure a standardised approach to the invasive procedures carried out within the critical care setting.

3. DEFINITIONS

3.1 **Insertion of a Central Venous Catheter (CVC) –** also known as a central line, central venous line, or central venous access catheter. This is a catheter placed into a large vein. Catheters can be placed in veins in the neck (internal jugular vein), chest (subclavian vein or axillary vein), groin (femoral vein), or through veins in the arms (also known as a PICC line, or peripherally inserted central catheters). These catheters are for gaining vascular access, to monitor central venous pressures and administration of intravenous medication.

3.2 **Insertion of chest drain –** this is the insertion of a chest tube (chest drain, thoracic catheter, tube thoracostomy, or intercostal drain) a flexible plastic tube that is inserted through the chest wall and into the pleural space or mediastinum. It is used to remove air (pneumothorax), fluid (pleural effusion, blood, chyle), or pus (empyema) from the intrathoracic space.

3.3 **Intubation –** Usually referred to as tracheal intubation – is the placement of a flexible plastic tube into the trachea (windpipe) to maintain an open airway or to serve as a conduit through which to administer certain drugs. It is frequently performed in critically injured, ill, or anaesthetised patients to facilitate ventilation of the lungs, including mechanical ventilation, and to prevent the possibility of asphyxiation or airway obstruction.

3.4 **Percutaneous tracheostomy**– Percutaneous tracheostomy also referred to as bedside tracheostomy, is the placement of a tracheostomy tube without direct surgical visualisation of the trachea. These are classically indicated to facilitate weaning in difficult to wean patients, to aid in tracheobronchial toileting, to protect airways in patients at risk of aspiration, in anticipated prolonged ventilator stay, and to minimise sedation requirements.

4. GOVERNANCE AND AUDIT

4.1 The Intensive Care Unit (ICU) Specialty Governance Group will monitor the implementation of this guideline and commission a bi-annual audit of adherence. See Section 3 and Appendix 5.

5. DOCUMENTATION OF INVASIVE PROCEDURES

5.1 A memorandum has been developed by the Faculty of Intensive Care Medicine of the Royal College of Anaesthetists for the procedures noted in Section 3 that have been developed for use when invasive procedures are undertaken in a critical care environment. The checklist incorporates the principles outlined in Sections 9, 10 and 11.

5.2 Appropriate consent will be gained by the responsible person for the procedure prior to all procedures and where possible discussed with patients.

5.3 If the procedures have been carried out within the operating theatre environment the procedure will be recorded within the patient's anaesthetic chart. This will detail the type of catheter and position.

5.4 If the procedures occur within the ICU it will be recorded within the patient's electronic notes on CareVue.

6. WORKFORCE, DUTIES AND RESPONSIBILITIES, TRAINING

6.1 Minimum workforce requirements for these invasive procedures are as follows:

- 1 x competent medical practitioner/1 x competent Advanced Critical Care Practitioner (ACCP)
- 1 x trained assistant and/or as required.

6.2 All staff without appropriate competency will be supervised.

6.3 It is the responsibility of each member of staff involved in the procedures outlined to:

- to comply with the standards set out in this guidance
- to work within their own competence
- to escalate any concerns regarding own and other's competence
- to report all incidents of non-compliance (including near miss events) using the Trust's Incident Reporting procedures on Datix (Incident Reporting, Analysing, Investigating and Learning Policy). Incidents will be discussed at the ICU and Anaesthetic/Theatres Governance Groups and any identified actions resulting from incidents will be monitored to implementation.

6.4 **Training**

All members of staff involved in the delivery of invasive procedures outlined in this guideline must complete either (1) the Trust's training as per Section 22 of the overarching Trust Safe Surgery and Interventional Procedures Policy or (2) the LocSSIP training presentation and verify their agreement to work and communicate as part of a team and to speak out if they have any concerns about patient safety.

7. SCHEDULING

7.1 These procedures will be performed as necessary when safe to do so in an appropriate environment.

8. HANDOVERS AND INFORMATION TRANSFER

8.1 **Pre-procedure**

Prior to the procedure, the information outlined on each individual Checklist (before the procedure) will be checked between staff and where appropriate with the patient to ensure that identity is confirmed, the patient understands the nature of the procedure, laterality, and that verbal consent is obtained

8.2 **Post-Procedure**

On completion of procedure the information outlined in each individual Checklist (before the procedure) will be checked between staff. The responsible person for the procedure will document the completion of the procedure confirming the checklist have been followed.

9. PROCEDURE VERIFICATION AND CONSENT

9.1 Where applicable, written or verbal consent is obtained in non-emergency situations and documented in the patient's notes in advance of the procedure. This is undertaken and recorded in line with the Trust's Consent to Treatment and Examination Policy

9.2 The consultant / doctor / practitioner will ensure that the correct site and side (if applicable) for the procedure is agreed at the pre-procedure safety check.

9.3 The patient's identity will be checked and confirmed with the patient using positive patient identification (involving the patient if possible).

9.4 Confirmation of the procedure will be verbalised by the practitioner and confirmed (involving patient if possible).

10. SIGN IN AND TIME OUT – COMBINED SAFETY CHECK

10.1 The team involved in the procedure will participate in the combined safety check as per the relevant checklist in the appendices. The following information should be confirmed and agreed by the team with the involvement of the patient where possible.

- Correct patient identity as per the Identification of Patients Policy
- Appropriate consent in place, or procedure documented as performed in patient's best interests
- Correct site identified (where applicable).

11. POST-PROCEDURE CHECKS

11.1 The team involved in the invasive procedure will participate in the Post-procedure checks as per the checklist in the appendices, ensuring the procedure is correctly recorded, and post-procedure management plan discussed and relevant information given to the patient.

12. DEBRIEFING

12.1 If any issues have arisen or problems such as equipment difficulties, a debriefing is to be undertaken and recorded by the responsible practitioner in the patient note, and a Datix completed if appropriate.

12.2 Any questions included at the debrief are professional and not personal and are designed to facilitate reflection, shared learning and make improvements where appropriate.

12.3 Examples of questions for discussion and consideration at debrief.

- Communication – any issues for improvement?
- Team harmony – for example, was the best use of skills utilised?
- Planning – were there any planning issues, ie missing stock, how can we improve?
- Equipment failure – any issues?
- Have any problems been resolved?
- Will anything need to be addressed prior to the next clinic / treatment session?
- Who is taking the responsibility to ensure actions?

12.4 All members of the procedural team should have noted any key points for consideration at the debrief that arose throughout the session. These should be raised and presented for discussion at the debrief.

12.5 All members present at debrief must feel comfortable enough to contribute to the debrief discussion and raise any concerns or questions.

13. MONITORING COMPLIANCE WITH THIS GUIDELINE

13.1 An audit will be undertaken by the ICU Audit Lead on a bi-annual basis to ensure that each of the procedures contained in this guideline is carried out in accordance with the guidelines and checklists in Appendices 1-4. Given the size of the unit and the fluctuation in the number and types of invasive procedures undertaken, this observational audit will be undertaken on a minimum of 3 of each type of procedure over a suitable course of time when the Audit Lead is undertaking Consultant of the Day duties.

13.2 Feedback will be given to (1) staff involved at the time the observational audit is completed, and (2) will be discussed at the first ICU governance meeting following completion of the requisite 12 audits.

14. ASSOCIATED CLINICAL GUIDELINES OR POLICIES/PROCEDURES

- Safe Surgery and Interventional Procedures Policy
- Consent to Treatment and Examination Policy
- Identification of Patients Policy
- Incident Reporting, Analysing, Investigating and Learning Policy and Procedures
- NG Tube Placement on ICU

APPENDIX 1

Central Line Insertion Checklist

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Before Procedure

- Assess bleeding risk
  - Coagulation results
  - Platelet count
  - Medications
  - Allergies
- Consent patient if awake
- Optimise position. Complete pre-procedure ultrasound scan.
- Equipment ready?
  - Run through pressure tranducer
  - Connect to monitor and zero
- Skin prep, Chloraprep, drape.

Time Out

- Standard monitoring visible?
  - Heart rate
  - Blood pressure
  - Saturations
- Check bed function and ability for head down position
- Correct line ready?
- How will venous cannulation be confirmed?
  - Pressure transduction
  - Venous gas
  - Ultrasound wire

IF ARTERIAL DILATATION OR CATHETERISATION* IS SUSPECTED

DO NOT REMOVE LINE

SEEK ASSISTANCE

* If arterial catheterisation with small bore introducer only – it can be removed and gentle pressure applied for 5 mins

After Procedure

- **HAS THE GUIDEWIRE BEEN REMOVED?**
- Confirm CVP wave form
- Secure line
  - Suture in place
  - Clear dressing
  - Caps placed in sterile fashion
- All lumens aspirated and flushed?
  - If you cannot aspirate the brown lumen, ensure guidewire has been removed
- Clean up
  - Dispose of sharps
  - Clean ultrasound machine
- Request chest x-ray
- Document insertion

This Checklist is not intended to be a comprehensive guide to central line insertion

Customised for Royal Devon & Exeter ICU/HDU
Approved as part of Critical Care LocSSIP: 28/11/18
Review Date: 28/05/21

APPENDIX 2

Chest Drain Insertion Checklist

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Before Procedure

- Consent patient if awake
- Team
  - Experienced operator?
  - Assistant throughout procedure
- Side confirmed?
- Imaging reviewed?
  - Chest x-ray, CT scan
- Bedside USS performed
  - For Seldinger drainage of effusions use real-time US
- Assess bleeding risk
  - Coagulation results
  - Platelet results
  - Medications / allergies
- Appropriate analgesia and sedation

Time out

- Position optimal?
- Standard monitoring visible?
- Drainage system ready to connect?
- Further help available?
  - ICU Consultant
  - ICU SpR
  - Obstetrics anaesthetist #981 (out of hours)

After Procedure

- **HAS THE GUIDEWIRE BEEN REMOVED?** *(if used)*
- Secure chest drain
  - Sutured and dressed?
  - Connected securely to drain bag or underwater seal?
  - Taps or clamps open to allow drainage?
- Clean up
  - Dispose of sharps
  - Clean ultrasound machine
- Review
  - ABG and ventilation
  - Sedation and analgesia
- Request chest x-ray
- Document procedure

This Checklist is not intended to be a comprehensive guide to chest drain insertion

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APPENDIX 3

Emergency Intubation Checklist

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Prepare Patient

- Can the patient's condition be optimised any further before intubation?
- Is there a previous laryngoscopy grade?
- Is the patient's position and pre-oxygenation optimal?
- How will anaesthesia be maintained after induction?
- Check allergies

Prepare Equipment

- What monitoring is applied?
  - ECG
  - Blood pressure
  - Sats probe
  - Capnography
- What equipment is checked and available?
  - Self inflating bag
  - Suction – on max
  - 2 ET Tubes
  - 2 laryngoscopes
  - Bougie
  - Tube tie/tape
- Do you have IV access and drugs required including vasopressors and fluid boluses?
- Is the ventilator checked and set up appropriately?

Prepare Team

- Who is...?
  - Team Leader
  - First intubator
  - Second intubator
  - Cricoid pressure
  - Intubator's assistant
  - Drugs
  - Manual in-line stabilisation (if indicated)
- How do we contact further help if required? (Bleep #981 out of hours)

Prepare for Difficulty

- If the airway is difficult, could we wake the patient up?
- If the intubation is difficult how will you maintain oxygenation?
- Is the C-MAC set up, and switched on?
- Where is the relevant equipment including alternative airway?
- Are any specific complications anticipated?

This Checklist is not intended to be a comprehensive guide to preparation for induction

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APPENDIX 4

Percutaneous Tracheostomy Insertion Checklist

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Before Procedure

Confirm:

- Indication
- Contraindications?
- Consent form/verbal assent/ family discussion
- Experienced team including nursing staff?
- FiO2 <50%, PEEP < 8cm H2O
- Laryngoscopy
  - Previous grade
  - Anaesthetics history checked
  - Difficult airway plan
- Bleeding risk
  - Plts >75/INR<1.5/APTT<45
  - Anticoagulation medications?
- Neck examined and USS considered?
- Optimal positioning and appropriate tracheostomy type?

Pre-Incision

- Adequate monitoring applied and visible?
- Adequately anaesthetised?
  - FiO2 100%/sedation/analgesia/ relaxant/vasopressor/IV access
- Enteral feed stopped, insulin held and NGT aspirated?
- Local anaesthesia infiltrated?

Time Out

- Team roles confirmed:
  - Anaesthetic and airway
  - Tracheostomy operator
  - Experienced nurse
  - Team leader
- Difficult airway trolley located?
- Further help available?
  - 2nd ICU Consultant
  - ENT SpR (via switch)

After Procedure

- Tracheostomy position confirmed with bronchoscopy **AND** E_tCO_2?
- Tracheostomy secured, inner tube inserted and connected to ventilator, cuff pressure checked?
- Chest x-ray ordered?
- Document procedure – include tracheal ring level.
- Blood gas and review of sedation, ventilation and feeding.
- Clean bronchoscope.

This Checklist is not intended to be a comprehensive guide to percutaneous tracheostomy insertion

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Appendix Critical Care LOCSSIP Observational Audit Tool

Auditor to confirm observation of critical elements contained in Checklist to ensure safe practice.

Central Line Insertion (Appendix 1)

- ☐ Assess bleeding risk inc meds/allergies
- ☐ Equipment ready
- ☐ Skin prep, Chloraprep, drape

- ☐ If arterial dilatation or catheterisation is suspected, do not remove line, seek assistance

- ☐ **HAS GUIDEWIRE BEEN REMOVED?**

- ☐ Confirm CVP wave form
- ☐ Secure line (suture, dressing, caps)
- ☐ Lumens aspirated and flushed?
- ☐ Clean up (sharps, US machine cleaned)
- ☐ Chest X-ray requested
- ☐ Insertion documented on CareVue

Date of Audit		Bed#:
Initials of operators		
Comments to feedback to staff and governance		

Emergency Intubation (Appendiendix 3)

- ☐ Is there a previous larygoscopy grade?
- ☐ How will anaesthesia be maintained after induction?
- ☐ Check allergies
- ☐ Correct monitoring applied (ECG, BP, sats, capnography)
- ☐ Equipment checked and available?

- ☐ IV access and drugs required including vasogressors and fluid boluses?
- ☐ Ventilator checked/set up appropriately
- ☐ If airway is difficult could we wake the patient
- ☐ If the intubation is difficult how will oxygenation be maintained?
- ☐ Is C-MAC set up and switched on?
- ☐ Where is relevant equipment incl. alt. airway?
- ☐ Are any specific complications anticipated?

Date of Audit		Bed#:
Initials of operators		
Comments to feedback to staff and governance		

Chest Drain Insertion (Appendix 2)

- ☐ Side confirmed
- ☐ Imaging reviewed
- ☐ Bedside US performed
- ☐ Assess bleeding risk inc meds/allergies
- ☐ Bedside US performed
- ☐ Equipment ready
- ☐ **HAS GUIDEWIRE BEEN REMOVED?**
- ☐ Chest drain secured (suture, dressing)
- ☐ Clean up (sharps, US machine cleaned)
- ☐ Review ABG and ventilation
- ☐ Review sedation and analgesia
- ☐ Chest X-ray requested
- ☐ Insertion documented on CareVue

Date of Audit		Bed#:
Initials of operators		
Comments to feedback to staff and governance		

Percutaneous Tracheostomy Insertion (Appx 4)

- ☐ Confirm indication
- ☐ Contraindications (FiO2 <50%, PEEP <8cm H2O)
- ☐ Laryngoscopy (previous grade, history, airway plan)
- ☐ Bleeding risk considered (anticoag meds?)
- ☐ Neck examined and US considered?
- ☐ Adequate monitoring applied and visible?
- ☐ Enteral feed stopped, insulin held, NGP aspirated?
- ☐ Local anaesthesia infiltrated?
- ☐ **Time Out**
- ☐ Difficult airway trolley located?
- ☐ Tracheostomy position confirmed
- ☐ Tracheostomy secured, connected, pressure checked
- ☐ Chest X-ray requested (if required)
- ☐ Insertion documented (inc tracheal ring level)
- ☐ Blood gas, review sedation, ventilation, feeding
- ☐ Clean bronchoscope

Date of Audit		Bed#:
Initials of operators		
Comments to feedback to staff and governance		